Bao Truong

+84 974634986 | giabaotruong.work@gmail.com

Bao Truong | giabao804 | Bao Truong

Research Interests

My current research lies at the intersection of computer vision and multimodal learning, with a specific focus on multi-sensory robotic perception. My primary objective is to develop foundational methods that enable robotic systems to perceive and act with human-like capabilities.

Education

- **Hanoi University of Science and Technology** — *Vietnam*
  BS in Automation and Control Engineering — 2021 - 2025
  CPA: 3.67/4.00 - Excellent Degree
- **Hatinh High School for Gifted Students** — *Vietnam*
  Gifted class in Mathematics — 2018 - 2021

Experience

- **AI Researcher** — ***Robotic Perceptions***
  FPT Software AI Center, Vietnam — May 2025 - present
  - Focusing on Multimodal Learning, Computer Vision and AI for Robotics.
- **AI Engineer** — ***Automatic Speech Recognition***
  VinBigData Joint Stock Company, Vietnam — July 2024 - May 2025
  - Building a framework for generating a multimodal (audio-language) datasets that served as the foundation for training the CallBot, ensuring it learns accurate conversation patterns, function callings, and appropriate responses within the customer domains.
  - Developing a framework with suitable design patterns for automatically evaluating ASR, Speech2Speech models.
- **Research Assistant** — ***Computer Vision***
  Image Processing and Signal Analysis Lab [], Vietnam — May 2023 - present
  - Medical Imaging, Signal Processing.

Selected Publications

[1]] **Bao Truong**, Quang Nguyen, Baoru Huang, Jinpei Han, Van Nguyen, Ngan Le, Minh-Tan Pham, Doan Duy Hien, Anh Nguyen:
"SIGMA: A Physics-Based Benchmark for Gas Chimney Understanding in Seismic Images".
CVPR 2026

[2]] **Gia-Bao Truong**, Thi-Thao Tran, Nhu-Linh Than, Van Quang Nguyen, Thi-Hue Nguyen, Van-Truong Pham:
"SC-MambaFew: Few-shot learning based on Mamba and selective spatial-channel attention for bearing fault diagnosis".
Computers and Electrical Engineering.

[3]] Viet-Thanh Nguyen, **Gia-Bao Truong**, Van-Truong Pham, Thi-Thao Tran:
"1S-MambaMatch: A semi-supervised and One-shot learning framework with Multi-input Visual State Space Model for skin lesion segmentation".
Biomedical Signal Processing and Control.

[4]] Van Quang Nguyen, Thi-Thao Tran, **Gia-Bao Truong**, Nhu-Linh Than, Van-Truong Pham:
"Dense Attention Mamba-based Network with Adaptive Sigmoid Fowlkes-Mallows Loss for Enhanced Medical Image Segmentation".
Image and Vision Computing.

Honors and Awards

- **First Prize in Scientific Research Student Award**
  The 42nd HUST Annual Student Research Conference in 2025
- **Academic Excellence Scholarship**
  HUST

References

1. **Assoc. Prof. Anh Nguyen**
   Department of Computer Science,
   University of Liverpool.
   Email: anh.nguyen@liverpool.ac.uk
   Bio; Google Scholar

2. **Assoc. Prof. Van-Truong Pham**
   Department of Control Engineering and Automation,
   School of Electrical and Electronic Engineering, Hanoi University of Science and Technology.
   Email: truong.phamvan@hust.edu.vn
   Bio; Google Scholar

3. **Assoc. Prof. Thi-Thao Tran**
   Department of Control Engineering and Automation,
   School of Electrical and Electronic Engineering, Hanoi University of Science and Technology.
   Email: thao.tranthi@hust.edu.vn
   Bio; Google Scholar